



SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme : B. E / B. tech

Course Code : CSA43

Course Name : Internet programming

Problem Statement (Assignment)

Max Marks: 100

Field	Content
Assignment Title	Smart Campus Service Management System
Course Outcome(s) – CO	CO1: Develop visually appealing web pages using CSS for layout and style, and create dynamic, interactive functionality with JavaScript. CO2: Build dynamic web applications using XML, XSLT, and JSP within the MVC paradigm for efficient data handling and presentation.
Bloom's Taxonomy Level	L3 – Apply L4 – Analyze L5 – Evaluate L6 – Create
SDG Mapping (SDG 1–17)	SDG 9 - Industry, Innovation and Infrastructure, SDG 4 - Quality Education

Problem Statement

A university currently relies on manual communication such as emails, phone calls, and paper-based records for reporting and resolving campus service issues. This approach makes it difficult to efficiently record complaints, assign requests to the appropriate service personnel, monitor progress, and provide timely status updates to students and faculty.

The university therefore proposes to develop a **Smart Campus Service Management System** as an **integrated web-based application**. The system shall enable students and faculty to submit service requests related to laboratory equipment faults, network problems, classroom maintenance, library services, and other technical issues through a web browser. Users should be able to provide relevant details, submit requests, and subsequently track the status of their requests.

The proposed system should demonstrate an integrated client-server web architecture. The client side shall provide an interactive and user-friendly interface using HTML/XHTML, CSS, JavaScript, and DOM. The server side shall process requests using Java Servlets and JSP, following the MVC architecture. XML shall be used for structured data representation, and XSLT shall be used to transform XML data for presentation where appropriate.

Problem Identification (10 Marks)-The university relies on manual communication (emails, phone calls, paper records) for reporting and resolving service issues, which causes inefficiency in recording complaints, assigning requests, monitoring progress, and providing timely updates.

Proposed Solution -Develop a **Smart Campus Service Management System (10 Marks)**, a web-based application that allows students and faculty to submit service requests (lab equipment faults, network problems, classroom maintenance, library services) and track their status online.

System Design & Architecture - Data Representation (20 Marks) -Implement an integrated **client-server web architecture**. The client side should use HTML/XHTML, CSS, JavaScript, and DOM for interactivity, while the server side should use Java Servlets and JSP following the **MVC model**. Use **XML** for structured data representation for transforming XML data into user-friendly formats for presentation and reporting.

Implementation & Testing (10 Marks) -Demonstrate request submission, assignment to service personnel, progress tracking, and timely updates. Validate inputs, handle errors, and ensure smooth user experience.

STG Goals –SDG 9 - Industry, Innovation and Infrastructure, **SDG 4** - Quality Education

Requirement Analysis (*SDG 9 – Industry, Innovation & Infrastructure*) **Identify inefficiencies manual reporting and propose a digital infrastructure for service management.**

Client Interface Development (*SDG 4 – Quality Education*) **Build user-friendly web forms and dashboards so students/faculty can easily submit and track requests, improving access to campus services.**

A. Assignment Problem

Challenge

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B. Problem Statement

The proposed system should demonstrate an integrated client-server web architecture. The client side shall provide an interactive and user-friendly interface using HTML/XHTML, CSS, JavaScript, and DOM. The server side shall process requests using Java Servlets and JSP, following the MVC architecture. XML shall be used for structured data representation, and XSLT shall be used to transform XML data for presentation where appropriate.

Problem Identification (10 Marks)-The university relies on manual communication (emails, phone calls, paper records) for reporting and resolving service issues, which causes inefficiency in recording complaints, assigning requests, monitoring progress, and providing timely updates.

Proposed Solution
-Develop a Smart Campus Service Management System (10 Marks), a web based application that allows students and faculty to submit service requests (lab equipment faults, network problems, classroom maintenance, library services) and track their status online.
System Design & Architecture
- Data Representation (20 Marks) -Implement an integrated client server web architecture. The client

side should use HTML/XHTML, CSS, JavaScript, and DOM for interactivity, while the server side should use Java Servlets and JSP following the MVC model. Use XML for structured data representation for transforming XML data into user friendly formats for presentation and reporting. Implementation & Testing (10 Marks) -Demonstrate request submission, assignment to service personnel, progress tracking, and timely updates. Validate inputs, handle errors, and ensure smooth user experience.

Requirements And Constraints:

Functional requirements

The smart campus service management system shall provide this functional requirements:

1. The system shall allow the student and faculty members to login with valid credentials.
2. The system shall allow the users to submit requests for campus service through a web browser.
3. The system shall provide service categories like laboratory equipment fault, network problem, classroom maintenance, library services, electrical technical issues, and other campus services.

The form made is:

1. Will you require a form to be submitted with the request?
2. Willing to concentrate people in some request.
3. Where interested parties will be located.
4. Requests priority, description of the problem, contact details of the person making the request.
5. The system shall generate a request ID for each submitted request.
6. The administrators shall be able to view all the submitted requests.
7. The administrators shall be able to assign requests to appropriate service personnel.
8. The service personnel shall be able to update progress of assigned requests.

A.3 Technical Constraints

These technical constraints apply to the proposed system:

- The client interface will be handled with HTML, HTMLX, HTML, CSS, JavaScript, and DOM.
- Java Servlets is used to response the request of the server side.
- Will use XML as persistent storage for structured data and eventually use XSLT whenever possible during XML data transformation.
- At least web browser/application should be run using a normal web browser.
- System should be deployable, install, and run over Java web application server e.g., Apache

Tomcat.

- Tests should be on both locations: server and on the client side.
- Control should be kept away from the control HTML presentation, so the presentation HTML
- won't be polluted with control HTML.
- The data/business logic should be separated from the presentation/control HTML, as well.

A.4 Cost and Time Constraints

The proposed system is designed as an academic web application and should preferably use freely available or open-source development tools. Development time is limited to the academic project schedule. Therefore, the initial implementation focuses on the essential features of request submission, assignment, tracking, and status updates rather than advanced enterprise-level functionality.

A.5 Resource Constraints

The system requires:

- A computer for development and testing.
- Java Development Kit (JDK).
- Apache Tomcat or another compatible Java web server.
- IDE such as Visual Studio Code, Eclipse, or IntelliJ IDEA.
- Modern web browser.
- XML and XSLT processing support.
- Basic server/database or structured XML storage depending on implementation.

A.6 Reliability Requirements

The system should be reliable in storing submitted requests and maintaining their current status.

Each request should have a unique request ID to prevent confusion between multiple complaints.

The system shall prevent unauthorized users from modifying requests that do not belong to them.

A.7 Sustainability and Environmental Considerations

The proposed system reduces dependency on paper-based complaint forms and physical records.

Digital submission and reporting reduces paper consumption and improves the efficiency of campus service operations.

Centralized digital records also reduces the need for repeated manual communication.

A.8 Security and Privacy Requirements

Security is an important requirement because the system may contain user information and service-related records.

The following measures should be considered:

- User authentication.
- Role-based authorization.
- Secure session management.

A.9 User Requirements

Students and faculty require a simple interface that allows them to:

- Submit complaints quickly.
- Select the appropriate service category.
- Describe their problem clearly.
- View assigned requests.
- Update progress.
- Record resolution details.

D.Student Work

1. Problem Understanding and Formulation

1.1 Problem Identification

The university currently utilizes manual means of communication like emails, phone calls and paper-based records to report campus service problems. Although such mechanisms are effective in individual complaint cases, they are not efficient when a high volume of requests is

received.

There is no central mechanism of registering complaints, assigning them to relevant personnel and tracking their progress. Additionally, students and faculty members may not be able to easily ascertain whether a complaint has been received, assigned, or resolved.

1.2 Expected Outcomes

The expected outcomes of the proposed system include:

- Faster service request submission by users.
- Centralized complaint registration.
- Proper allocation of requests.

1.3 Available Information and Data

The system will deal with the following information:

- User ID
- User name
- User role
- Service request ID
- Request category

The above information can be stored in a structured format using XML. Additionally, the data can be stored in a suitable persistent data store.

1.4 Assumptions

The following assumptions are made:

The users have access to web browsers, university network/internet

The users provide accurate information while submitting requests

The administrators are the ones who assign requests

The service personnel have the necessary permissions to update allocated service requests

1.5 Constraints

The major constraints of the project include:

- Limited academic development time
- Limited development resources
- Use of Java Servlets and JSP

2. Application of Course Knowledge

The project draws on various concepts from Internet Programming and Web Application Development.

2.1 Client-Server Architecture

The application is built on the client-server architecture. The client side is the web browser through which students, faculty members, administrators, and service personnel access the application.

The overall working principle is as follows:

Client Browser → HTTP Request → Web Server → Servlet → Model/Business Logic → JSP/XML/XSLT → HTTP Response → Client Browser

2.2 HTML/XHTML

This project relies heavily on HTML/XHTML for building the application. HTML/XHTML forms are used to collect service requests from users. Some interface elements that rely on HTML/XHTML include:

- Login Form
- Service Request Form
- Request Tracking

2.3 CSS

Cascading Style Sheets (CSS) are employed to add styling to the application. It offers the following capabilities:

- Consistent font
- Visual spacing
- Form styles
- Menus

2.4 JavaScript and DOM

JavaScript brings interactivity to the web application. The Document Object Model (DOM)

enables the interaction between the web page and JavaScript. With JavaScript, it is possible to:

- Validate data
- Show messages
- Respond to mouse events

2.5 Java Servlets

Java servlets mainly act as controllers.

2.6 JSP

Java Server Pages (JSP) are used to produce web pages for the application. JSP can present data that servlets have processed. Examples of pages that use JSP:

- User Dashboard
- Status page for service request

2.7 MVC Architecture

The project adopts the Model View Controller (MVC) architectural pattern. This means the application is split into three main components:

Model → This is where service request data is stored and managed.

View → This consists of HTML/XHTML, CSS, and JavaScript.

Controller → This consists of Java servlets.

This architecture keeps the code organized and easy to maintain.

2.8 XML

The Extensible Markup Language (XML) is used for storing service request data. It is a markup language similar to HTML.

An example of XML:

- SCR1001
- Network Problem
- Block A – Lab 2

2.9 XSLT

Extensible Stylesheet Language Transformations (XSLT) can be used to convert XML data into a more readable format. For example, a list of service requests in XML can be turned into an

HTML table using XSLT. The following example shows how XSLT helps separate data from presentation:

- XML Code
- SCR1001
- Network Problem
- Block A – Lab 2
- High
- In Progress
- Network Support Team

2.10 HTTP Request-response Model

The application operates on the HTTP request-response model. This means the client browser sends an HTTP request to the server, and the server replies with an HTTP response.

Example

User fills the service request form → Browser sends HTTP post request to servlet → Servlet handles request → Servlet validates data → Servlet processes request → Servlet forwards response to JSP page → JSP page responds to browser with confirmation or updated status.

3.DESIGN

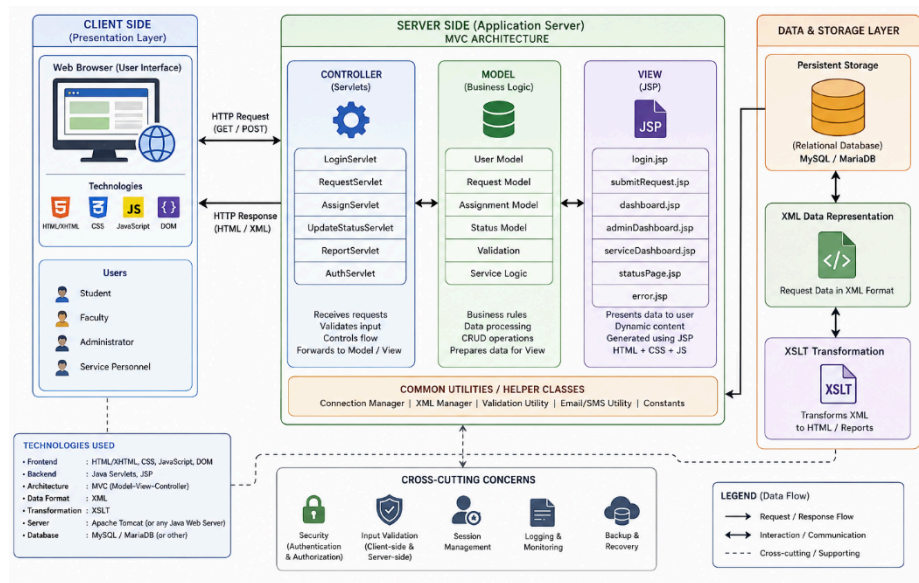
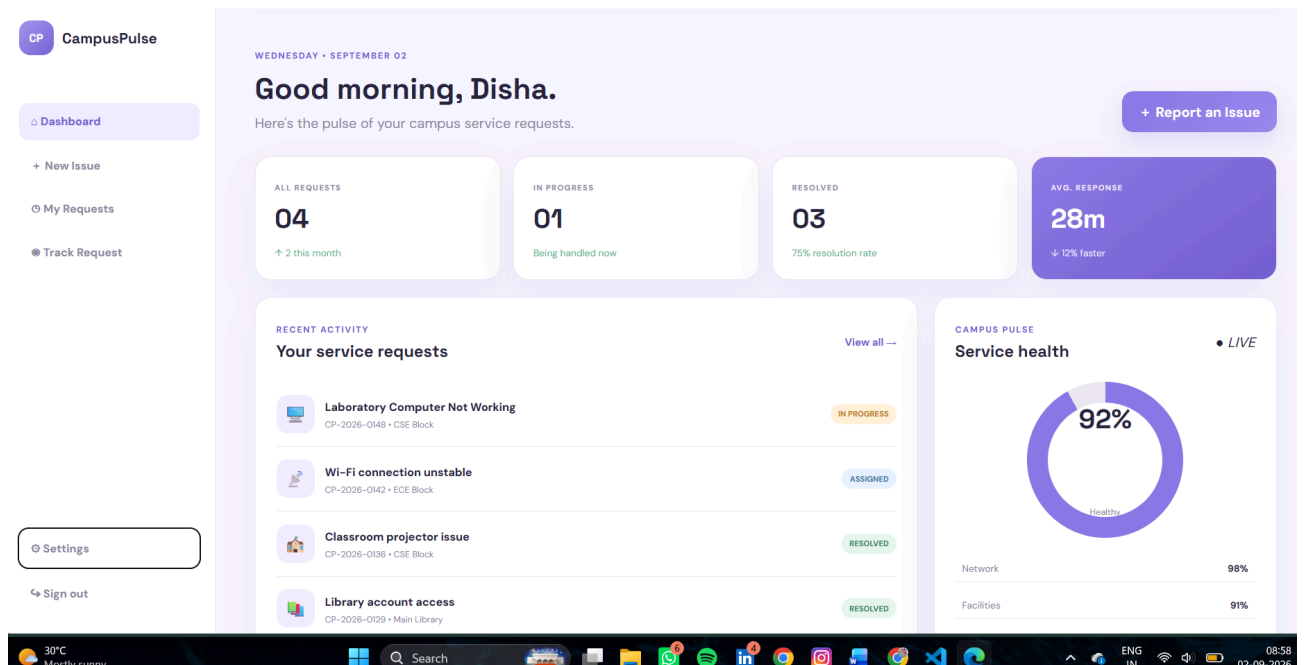
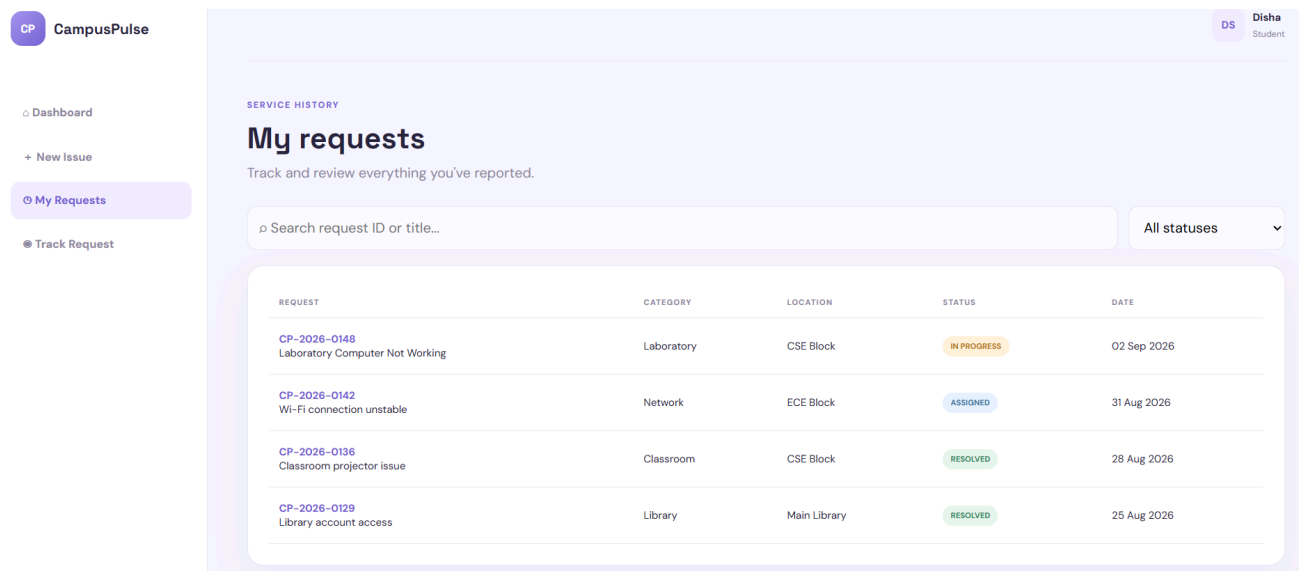


Fig:1 System Architecture

Use of Modern Tools





I have used HTML to develop these webpages.

3. Results and Validation

CP

CampusPulse

Dashboard

New Issue

My Requests

Track Request

SERVICE HISTORY

My requests

Track and review everything you've reported.

Search request ID or title...

All statuses

REQUEST	CATEGORY	LOCATION	STATUS	DATE
CP-2026-1626 Computer Lab is not Working ...	Laboratory	undefined	SUBMITTED	02 Sep 2026
CP-2026-0148 Laboratory Computer Not Working	Laboratory	CSE Block	IN PROGRESS	02 Sep 2026
CP-2026-0142 Wi-Fi connection unstable	Network	ECE Block	ASSIGNED	31 Aug 2026
CP-2026-0136 Classroom projector issue	Classroom	CSE Block	RESOLVED	28 Aug 2026
CP-2026-0129 Library account access	Library	Main Library	RESOLVED	25 Aug 2026

CP

CampusPulse

Dashboard

New Issue

My Requests

Track Request

LIVE TRACKING

Request #CP-2026-0148

Laboratory Computer Not Working

IN PROGRESS

Laboratory Computer Not Working

CSE Block • Central Lab • High priority

✓ Request submitted

02 Sep 2026 • 08:15 AM

✓ Request verified

02 Sep 2026 • 08:21 AM

• Technician assigned

Arun Kumar • Technical Support

○ Repair in progress

Waiting for technician update

○ Resolved

Pending

ASSIGNED TEAM

AK

Arun Kumar

Technical Support

Priority

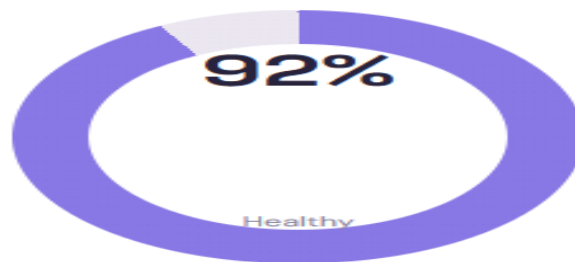
Submitted

Response ETA

High

08:15 AM

25 min



Network	98%
Facilities	91%
Library	94%

4. Analysis and Engineering Decision

The candidate solution of Campus Service Issue Reporting and Management System was evaluated as an alternate solution to the university's current manual complaint-registration and management system. As can be seen from the analysis, it would be more efficient, reliable, and transparent than using email, phone calls, and paper records.

- **Comparison of Alternatives**

The current method requires users to communicate their complaints and issues using a combination of communication channels. The proposed solution, on the other hand, would allow students and faculty to report them using a web-based interface. It would also allow the service personnel to view, assign, and manage complaints.

- **Manual System Proposed System**

- Complaint registration Email, phone calls, or paper records Online form
- Record maintenance Paper or electronic files Centralized database
- Assignment Manual based on communication From the system
- Status updates N/A Real-time

Advantages

The proposed system would be more reliable than the current manual one because complaints would be stored electronically and be easier to organize and manage. They could also be sorted

according to the type of service needed and assigned to the appropriate personnel.

Limitations

The proposed solution requires users to have access to a computer or smartphone and the Internet. It also has a steeper learning curve than simply calling or emailing the administration.

Trade-offs

The main trade-off is between the convenience of the current system and the efficiency of the proposed one. The current manual system requires no special tools or training to operate but has limited scalability and can become unwieldy as the number of complaints increases. The proposed system, however, has to be designed and implemented but will be more scalable and reduce the administrative burden significantly.

SDG Goals Related to the Project

SDG 9 – Industry, Innovation and Infrastructure

- Identify inefficiencies in manual campus service reporting.
- Replace paper-based complaint reporting with a digital service management system.
- Provide a centralized platform for submitting and managing campus service requests.
- Improve communication between students, faculty, administrators, and service personnel.
- Support digital infrastructure for efficient campus service operations.
- Enable systematic assignment and tracking of service requests.

SDG 4 – Quality Education

- Build user-friendly web forms for students and faculty.
- Provide easy access to campus service requests through a web browser.
- Allow users to track the status and progress of submitted requests.
- Reduce delays in resolving classroom, laboratory, network, library, and electrical issues.
- Improve the overall campus environment for students and faculty.
- Support efficient access to essential campus services.

Project Objectives

- Develop a web-based campus service management system.
- Allow students and faculty to submit service requests online.

- Generate a unique request ID for every complaint.
- Allow administrators to view and assign requests.
- Allow service personnel to update request progress.
- Maintain centralized digital records using XML.
- Improve the efficiency, reliability, and transparency of campus service management.

Final Solution Selection

The Campus Service Issue Reporting and Management System is the preferred solution because it directly addresses the issues with the manual one. As an information system, it would allow data to be stored, managed, and retrieved more efficiently, and reduce the administrative workload. Its features would allow students and faculty to report and track their complaints, and the appropriate personnel to manage and monitor them.

Quantitative and Qualitative Justification

The evaluation criteria can be measured in terms of the time required to process, assign, and resolve a complaint, the number of unresolved complaints, and user satisfaction. For example, if a student has to spend several minutes on the phone or via email describing their issue and providing personal information, the online form would reduce the time spent on registration.

5. Broader Considerations

The proposed Campus Service Issue Reporting and Management system will have broader considerations beyond addressing the issues of complaint registration and resolution in the university. The system will ensure sustainability, environmental friendliness, societal relevance, safety, ethics, and cost-effectiveness, thereby improving campus operations in the long term.

● Sustainability

The system will be sustainable because it will digitize the complaint registration and management process instead of relying on manual search and record-keeping systems. Complaints, assignments, and updates will be stored digitally and retrieved as needed.

- Environment

The proposed solution will be environmentally friendly because it will minimize paper usage in the complaint management system. The system will also reduce the storage of physical documents, as information will be stored digitally and retrieved whenever necessary.

- Society

The system will meet the societal needs of the university community by ensuring seamless communication among staff, students, and service personnel. For instance, complaint registration will be easy because students and staff will not need to provide personal contacts, such as phone numbers, to submit complaint details.

- Safety

The system will enhance campus safety by helping management monitor and respond to complaints about defects in electrical appliances, water, lights, and other service issues. For example, staff can set urgency levels for complaints based on the complexity of the problem and the resources required to address them.

- Ethics

The ethical considerations of the proposed solution relate to the security of information in the system and the protection of the complainant's privacy. Information in the system should be accessible only to authorized users after verifying their credentials. The complaint management system should also be designed to limit access to certain information.

- Economics

The proposed system will be cost-effective because it will reduce expenses for purchasing paper and printing complaint forms, minimizing the need for hard-copy records. Although the initial investment in system development may be high, the university will save on maintenance costs due to the elimination of manual search and record systems.

- Accessibility

Accessibility concerns the design and ease of use of the proposed system. For instance, the university can design the system to enable students and staff to navigate the complaint registration and management process easily.

6. Conclusion

Proposed Solution

The proposed solution is a Campus Service Issue Reporting and Management System. The solution uses technology to manage complaint registration, duty assignment, monitoring, communication, and record-keeping, replacing the current manual approach that relies on emails, phone calls, and paper records.

Major Findings

The findings show that the manual approach is less efficient than a centralized online system for handling campus service issues. The current approach has various limitations that hinder effective

complaint registration, appropriate duty allocation, effective communication, information recording, and data retrieval when needed.

● Achievement of Requirements

The proposed system will meet the requirements of the current situation as outlined below:

1. The system will provide an online portal for registering campus-related service issues.
2. The system will store complaints centrally and securely, reducing the risk of losing records.

● Limitations

The proposed system has several shortcomings, as discussed below:

1. The system requires an internet connection and servers to operate.
2. The installation, deployment, and operation of the system may be costly.

● Possible Improvements

Several improvements can be incorporated to make the system more effective and efficient in achieving its objectives. These modifications include:

1. The system can be configured to send automated emails or text messages to registered users, notifying them of their complaint's progress.
2. The system can be configured to allow campus security or maintenance teams to respond to emergencies such as electrical faults, water leakages, and other urgent service-related issues.

7. References

1. United Nations Department of Economic and Social Affairs (UN DESA). (2026). Goal 9: Industry, innovation and infrastructure.
2. This source is relevant to your requirement analysis as it emphasizes sustainable, reliable infrastructure, technological development, innovation, and information and communication technology.
3. UN SDG 9 – Industry, innovation and infrastructure

4. United Nations Department of Economic and Social Affairs (UN DESA). (2026). Goal
This source connects to your project as it addresses the importance of reliable infrastructure, innovation, digital transformation, skills, and enabling systems, which are key aspects of your proposed digital infrastructure for campus service management.

Assessment Criterion	CO Mapping	Weightage / Marks
Problem understanding & formulation	CO1	10
Application of course/domain knowledge	CO2	20
Solution methodology / design / implementation	CO3	20
Use of appropriate modern tools / techniques	CO4	10
Results, testing & validation	CO4 / CO5	15
Analysis, trade-offs & justification	CO5	15
Broader considerations / professional responsibility, where	CO5	5

Assessment Criterion	CO Mapping	Weightage / Marks
applicable		
Technical documentation & reflection	CO1–CO5	5
TOTAL		100

Performance Level	CO Attainment Percentage	Interpretation
Level 3	$\geq 70\%$	CO Attained
Level 2	60–69%	CO Moderately Attained
Level 1	50–59%	CO Minimally Attained
Level 0	$< 50\%$	CO Not Attained

Code of Conduct

I Disha P(192411086) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic-integrity rules will result in disciplinary action.

Signature of the Student: Disha P

_____ Faculty In-charge Name: Date:	_____ Course Coordinator Name: Date:	_____ Head of Department Name: Date:
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